Question 1:

Define what it means for a function a multi-variate function f to be symetric.

Answer:

A function $f(x_1, x_2, \dots, x_n)$ is said to be symmetric, if:

$$\forall \pi \in S_n : \quad f(x_1, x_2, \dots, x_n) = f(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(n)})$$

Where, S_n is the set of all permtuation operators on n elements.

Question 2:

Show that for any symetric polynomial, say p, were can write p in the following form

$$p(x_1, \dots, x_n) = a_1 V_1 + \dots a_n V_n$$

Where $V_k = \sum_A \prod_j^k x_j$, i.e. the sum of all monomolias of degree k (their permutation)

Answer:

Proof:

$$f_{imp}(x_1, x_2) = f_{or}(f_{not}(x_1))x_2$$

$$= f_{not}(x_1) + x_2 - f_{not}(x_1)x_2$$
$$= (1 - x_1) + x_2 - (1 - x_1)x_2$$

$$= 1 - x_1 + x_2 - (x_2 - x_1x_2)$$

$$= 1 - x_1 + x_2 - x_2 + x_1x_2$$

$$= 1 - x_1 + x_1x_2$$

The coordinate vector of f with respect to the ordered basis $\{1, x_1, x_2, x_1x_2\}$ is

$$[f]_B = (1, -1, 0, 1)$$

Now Consider what happens if we permute the inputs into f. That is we want $f(x_2, x_1)$ instead of $f(x_1, x_2)$

Then

$$f(x_2, x_1) = 1 - x_2 + x_1x_2$$

And the coordinate vector in the B basis is

$$[f]_B = (1, 0, -1, 1)$$

$$E_{(x_1,\ldots,x_n \in bern^n(\frac{\mu}{n}))}[\langle f, x\rangle]$$

$$= \sum fPr(\vec{x})$$

$$= \sum f(x)(\frac{\mu}{n})^n$$

$$= \sum \langle f, \vec{x}\rangle(\frac{\mu}{n})^n$$

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Question 3:

Define inclusion map

Answer:

Let A and B be sets, with $A \subset B$. Then we define an *inclusion map* as the map $i\colon A \to B$ where

$$i : a \in A \mapsto a \in B$$

That is, i takes an element in A and takes it into the same element in B. An example is $\mathbb{Z} \to \mathbb{R}$ $i(3 \in \mathbb{Z}) = 3 \in \mathbb{R}$

Question 4:

We claim that the map $\psi : \mathbb{Z} \to R$ is unique and given by $\psi : n \in\mapsto \sum^n 1_R \in R$.

Show why the map $\Phi : \mathbb{Z} \to R$ given by $\Phi\colon n \in \mathbb{Z} \mapsto 0_R^n \in R$, does satify.

Answer:

[below is wrong!!!]

By counter example, consider Φ on $1+2 \in Z$. That is consider $\Phi(1+2) = \Phi(3) = 0_R^3$

On the other hand, we have

$$\Phi(1) + \Phi(2) = 0_R^1 + 0_R^2$$

Using R's distributive axiom

$$= 0_R^1(1_R + 0_R)$$

Using the fact that 0_R is R's additive identity

$$= 0_R^1(1_R)$$

Using the act that 1_R is R's multiplicative idenitty.

$$= 1_R$$

Indeed $0_R^3 \neq 1_R$

Question 5:

In exact case, what is it that we want?

Answer:

Let $(x_1, \dots, x_m) \in [0, n]^m$

1. $\exists x_i = 0$, then output 0
2. $\forall x_i \neq 0$, then output 1

This is AND composition OR.